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Review

Recent applications of AI to environmental disciplines: A review

Aniko Konya [a,*], Peyman Nematzadeh [b]

[a] University of Illinois, Chicago, IL 60637, USA
[b] Independent Researcher, Greater Chicago Area, IL 60527, USA

HIGHLIGHTS

- We reviewed the application of artificial intelligence in the environmental field.
- More than 65 % of environmental tasks that utilize AI employ mathematical models.
- AI tools can greatly assist and benefit all environmental professionals.
- AI model training produces carbon emissions, but AI is still in early development.
- Environmental and AI professionals must collaborate to train new experts.

GRAPHICAL ABSTRACT



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ABSTRACT

The rapid development and efficiency of Artificial Intelligence (AI) tools have made them increasingly popular in various fields and research domains. The environmental discipline is now experiencing an exponential interest in harnessing the potential of AI over the past decade. We have reviewed the latest applications of AI tools in the environmental disciplines, highlighting the opportunities they present and discussing their advantages and disadvantages in this field. After the emergence of deep learning algorithms in 2010, interest in using AI tools for environmental tasks has grown exponentially. Among the studied articles, over 65 % of environmental tasks that demonstrate interest in using AI tools initially relied on conventional statistical and mathematical models. Using AI tools can greatly benefit the areas of environmental science and engineering. One of the main advantages of utilizing AI tools is their ability to analyze and process large amounts of data efficiently. Recently, the European Union established a European supercomputing ecosystem program to advance science and enhance the quality of life for its citizens. Nine of these projects prioritize environmental and sustainable goals. Despite the benefits of AI, it is still in its early stages of development, which comes with environmental concerns. The amount of power consumed and the time required to train an AI model can greatly affect the carbon emissions it produces, exacerbating the challenges posed by climate change. Efforts are currently underway to develop AI technology that is environmentally sustainable, minimizes energy consumption, and has a low carbon footprint. Selecting the appropriate AI model architecture can reduce energy consumption by almost 90 %. The main finding suggests

* Corresponding author at: 909 S Wolcott Ave, Chicago, IL 60612, USA.
E-mail address: akonya@uic.edu (A. Konya).

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that collaboration between environmental and AI professionals becomes crucial in leveraging the full potential of AI in addressing pressing environmental challenges.

1. Introduction

Many computer scientists, engineers, mathematicians, and psychologists have attempted to define the term "Artificial Intelligence (AI)" (McCarthy et al., 1955; Russell and Norvig, 2010). Despite these attempts, the definition of AI is still being standardized (European Commission, 2020). McCarthy et al. (1955) defined it as "the science and engineering of making intelligent machines, especially intelligent computer programs, … to achieve goals." These goals can range from playing chess to using a search engine or driving an autonomous car. AI is a universal discipline and can be involved in numerous intellectual tasks (Russell and Norvig, 2010). AI applications are not limited to the computer science field but also extend to engineering, mathematics, physics, astronomy, robotics, medicine, molecular biology and genetics, chemistry, transportation, digital marketing, social sciences, forensics, and precision agriculture. AI tools have become increasingly popular in various fields and research due to their fast development and efficient solutions. AI tools offer many professional and personal benefits in many disciplines, including environmental science and engineering. Despite being inactive in the environmental field for decades, recent studies reveal a growing interest in using AI tools in environmental science and engineering (Haupt et al., 2022).

The complexity of environmental data is a challenging obstacle for professionals to overcome. Fortunately, innovative computer programs provide us with a solution for achieving precise predictions, constant monitoring, risk evaluations, and other valuable advantages (Haupt et al., 2022). AI techniques possess the ability to reason, make decisions, and learn from data. They can transform real-world information into machine-understandable and usable knowledge and make decisions through planned pathways of optimization and solution searching (European Commission, 2020). Machines can learn from data, extract information, solve problems, adapt to changes, and predict behavior. One area of AI that focuses on learning and perception is Machine Learning (ML) and Computer Vision (Molnar, 2023). Both techniques are valuable in managing intricate data, like environmental data. For example, ML classifiers can detect and identify illegal landfills from unmanned aerial vehicle (UAV) images and videos, making territory scanning more efficient and cost-effective (Torres and Fraternali, 2021). Nature conservation tasks can also benefit from machine learning algorithms, which can help map habitats, model species distribution, and monitor forest health. Additionally, ML models can assist in decision-making processes for creating no-take zones and implementing protective measures to preserve local biodiversity (Da Silveira et al., 2021).

AI tools can also assess even more complex environmental data, such as predicting spatiotemporal environmental data (Amato et al., 2020) or evaluating and forecasting water quality (Georgescu et al., 2023). Deep learning (DL), a sub-field of machine learning (ML), is an effective technique for processing and analyzing intricate data. It uses neural networks with specialized layers to extract deeper levels of information at each layer (Haupt et al., 2022). To understand and interpret environmental data, various geophysical parameters need to be processed, like physical, chemical, and biological properties of surface waters, atmosphere, and land surface. These parameters can be measured in several ways, such as vegetation coverage, moisture content, species concentration, aerosol concentration, and more (Haupt et al., 2009). Researchers are developing deep learning algorithms that can comprehend information collected by different sensors. These algorithms are leading to significant advancements in environmental research, including climate modeling, energy conservation, decision-making, and much more (European Commission, 2022).

Environmental professionals have shown increased interest in using

Table 1
A comparison of recent reviews on the use of AI in environmental disciplines was made, highlighting their features in relation to the current review.

Title	Objective	Specific aims/ findings	Authors
Artificial intelligence and machine learning tools for high-performance microalgal wastewater treatment and algal biorefinery: A critical review	Reviewed the most promising AI/ML algorithms that has a potential to be applied in microalgal technologies in water treatment.	Commonly used techniques are artificial neural networks, support vector machine, genetic algorithms, decision tree, and random forest algorithms.	Oruganti et al., 2023
The History and Practice of AI in the Environmental Sciences	The evolution of AI applied to atmospheric science. From the American Meteorological Society (AMS) Committee on Artificial Intelligence Applications to Environmental Science (AI conference) perspective.	Reviewed the early applications (from 1998) of AI methods used for works presented in AMS AI conference, special focusing on atmospheric sciences.	Haupt et al., 2022
Artificial intelligence applications for sustainable solid waste management practices in Australia: A systematic review	AI technologies were examined to enhance sustainable waste management practices in Australia.	Compared to other models used in forecasting solid waste generation and recycling, AI-based models have better prediction abilities.	Andeobu et al., 2022
Sustainable AI: AI for sustainability and the sustainability of AI	Proposed the definition of Sustainable AI	The paper engaged with the three pillars of sustainability: social, economic, and environmental perspective.	Van Wynsberghe, 2021
Artificial intelligence in the water domain: Opportunities for responsible use	Reviewed current AI applications in the water domain, with a specific emphasis on responsible AI.	Suggested collaboration between data scientists and water professionals should develop responsible AI techniques for the water sector, with support from social sciences and humanities.	Doorn, 2021
A Review of Artificial Intelligence Applications to Achieve Water-related Sustainable Development Goals	Current application of AI in the water sector	Predictive maintenance of water infrastructure, forecasting water demand and consumption, water monitoring of reservoirs, and monitoring water-related disasters.	Mehmood et al., 2020

Artificial Intelligence (AI) in their work, especially since 2020 (Table 1). Earlier reviews (Oruganti et al., 2023; Haupt et al., 2022; Andeobu et al., 2022; Van Wynsberghe, 2021; Doorn, 2021; Mehmood et al., 2020) focused mainly on the use of AI in water management, earth observation, sustainability, waste management, or discussing the history of AI. These studies have primarily looked at data collection and processing techniques. We present our findings based on the environmental tasks where AI is applied. We would like to encourage all environmental professionals to discuss and explore how AI tools can benefit their work. This paper explores the current and potential uses of AI in environmental science and engineering fields. We examine how AI applications can positively impact the environment while also acknowledging any potential drawbacks in the fields of environmental science and engineering. Through our scoping review, we analyze the latest applications of AI tools in the environmental discipline and weigh their advantages and disadvantages. Our objective is to initiate a discussion among environmental professionals about this emerging topic and highlight the valuable contribution that AI can make. Additionally, we aim to encourage collaboration between environmental and AI experts, training a new generation of professionals to develop innovative approaches in this field.

2. AI applications in the environmental disciplines

Environmental science and engineering tackle complex issues that arise from different natural environments, including soil, water, air, organisms, and biological data. Thanks to current AI technologies, such a large amount of data can be efficiently managed and assessed. Gathering data from various sources, including demographic information, traffic trends, energy usage, and socioeconomic factors, can provide valuable insights and help predict environmental conditions. Integrating big data and artificial intelligence offers new possibilities for different environmental tasks such as modeling, monitoring, and research (Haupt et al., 2009). There are ongoing efforts to develop AI tools to address various environmental challenges. The potential for AI to improve environmental management and sustainability is vast, and we can expect to see even more innovative solutions in the future (European Commission, 2022). The field of artificial intelligence has made notable advancements lately, already benefiting numerous fields including medicine, microbiology, precision agriculture, and more. It is important to understand the current state of AI tools in fields like environmental science and engineering, which also deal with complex and big data. In this article, we will thoroughly examine the diverse applications of AI tools in the field of environmental science and engineering. We will elaborate on various notable examples where AI has enhanced and advanced this area of expertise to unprecedented levels.

We carried out a thorough scoping review using a systematic and iterative approach to identify and synthesize recent and emerging literature on the utilization of AI tools in the environmental disciplines. We searched extensively across ScienceDirect, Semantic Scholar, and Google Scholar to locate relevant studies covering various topics such as AI applications, machine and deep learning, computer vision, and different environmental disciplines. We focused on articles published between 2000 and 2023 and screened their titles and abstracts to determine their relevance to AI application in environmental science and engineering. To avoid counting duplicate articles gathered from the search engines, we summed and averaged the total number of relevant articles produced during this time period. The information we have gathered is an estimate and can be viewed in Figs. 1 and 2.

2.1. A brief history of AI applications in the environmental fields

Over the past two decades, between 2000 and 2023, there has been an increase in the use of AI applications in environmental science and engineering. As shown in Fig. 1, most (above 65 %) AI tools have been used for environmental engineering tasks, environmental monitoring, energy analysis, research, and predicting weather patterns. These tasks involve using traditional statistics and mathematical models to solve various issues. For example, when investigating greenhouse gas emissions for product life cycle assessments (Huang et al., 2023) or

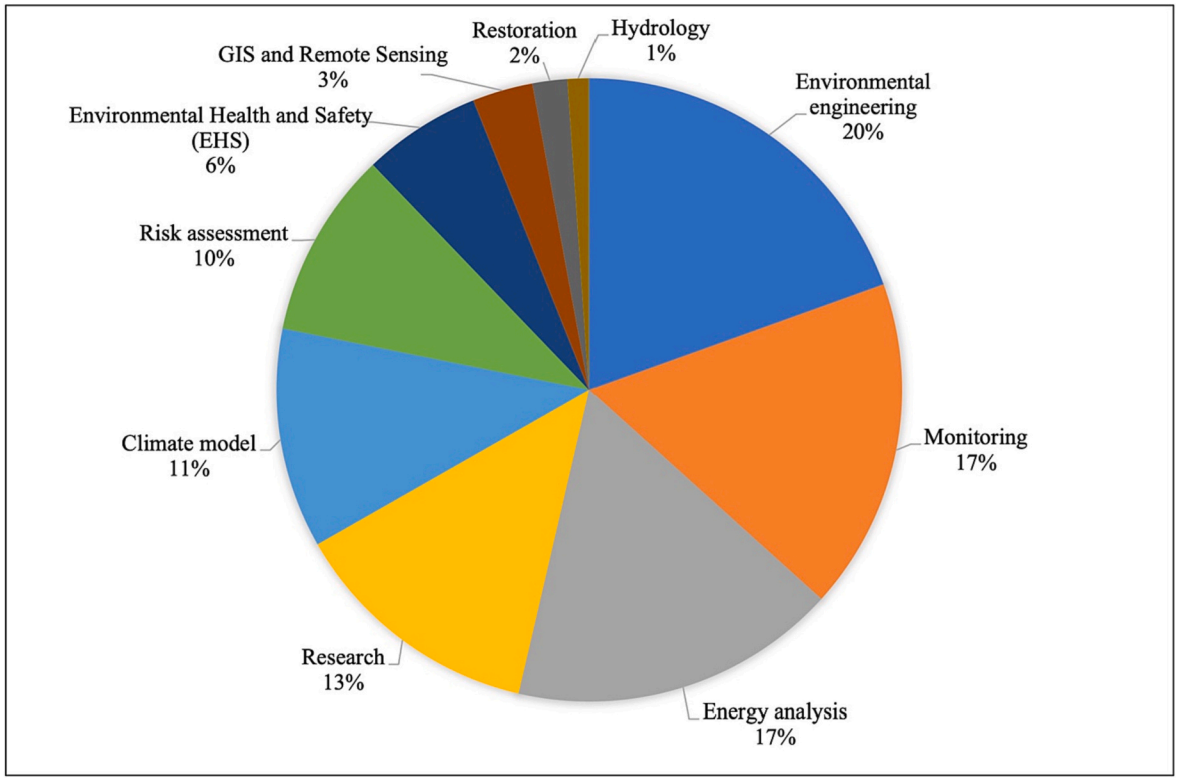


Fig. 1. The environmental disciplines where AI tools are regularly applied between 2000 and 2022.

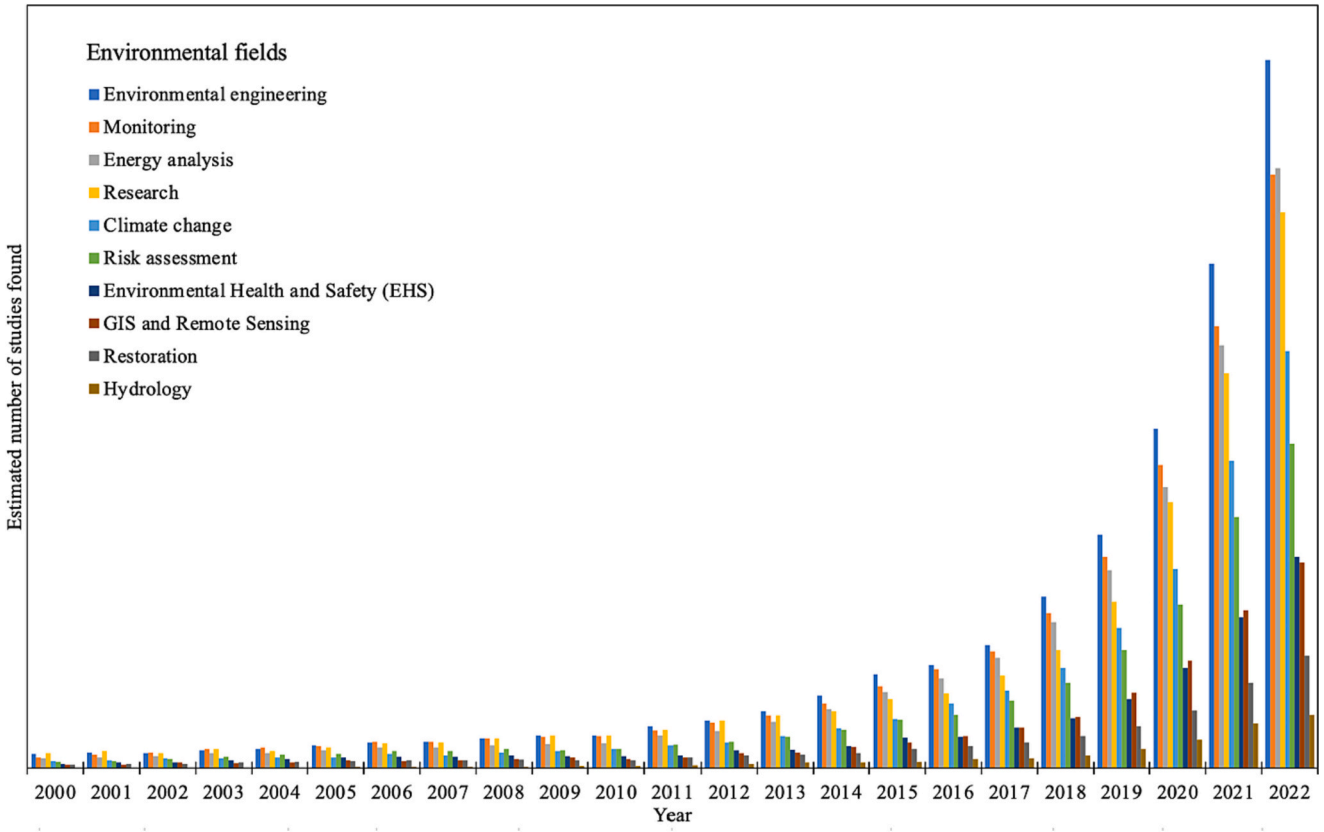


Fig. 2. Interest in using AI tools has seen a significant increase in environmental disciplines in recent years.

sustainable building materials for urban development (Bošković and Radivojević, 2023), sensitivity analysis is conducted based on highly complex mathematical models. Manually implementing these models would be challenging, requiring a lot of time and effort, and prone to errors. Machine learning (ML) algorithms have become a valuable tool for environmental professionals to overcome these difficulties. The use of ML in the environmental discipline became more common after 2000, with an increased interest rate with the emerging deep learning (DL) algorithms in 2010 (Haupt et al., 2022). This has made machine learning a crucial foundation for various environmental software services and applications (Gomes de Sousa et al., 2019). Environmental professionals began incorporating AI tools into their work in the early 2000s. However, it was after 2010, when DL algorithms emerged, that the usage of these tools saw an exponential increase, as shown in Fig. 2.

2.1.1. *Earlier environmental applications of AI tools*

The rapid expansion of AI technologies can be attributed to advancements in hardware design, which have enabled greater memory capacity and faster, more efficient computation. This progress has greatly enhanced machine and deep learning over the last decade (Marcus and Davis, 2019). Before the advent of machine learning, knowledge-based systems, a category of computer programs designed for problem-solving, were used for environmental assessment and decision-making (Geraghty, 2002). In the early 2000s, Bonano et al. (2000) utilized a single framework that combined risk assessment and decision analysis prototype techniques to evaluate, prioritize, and select preferred remediation alternatives for a contaminated site in the United States. They encountered a challenge in setting up multiple performance standards and deciding on suitable ranges of values that could be practically assessed for each impact measure. In the mid-2000s, Cetiner et al. (2005) used field experimental data to train Artificial Neural Networks (ANN). The goal was to create a mapping system that could link measurable features such as environmental conditions, input and output water temperatures, solar radiation, and flow rate of hot water. The intention was to use this mapping system for generating hot water using solar energy. The system displayed a 40 % increase in efficiency, with the highest power output at noon and the lowest in the afternoon.

After expert systems, the field of machine learning has experienced a significant evolution over the years. This has gradually led to the development of deep learning techniques, which have revolutionized the way we tackle complex problems using computers (Marcus and Davis, 2019). In the late 2000s, a method for real-time anomaly detection in environmental data streams was used by Hill and Minsker (2010) to identify measurement errors in windspeed data from Corpus Christi, Texas. The model used data to predict intervals for the data stream, allowing efficient handling of large amounts without classifying anomalies beforehand. The data was affected by errors lasting from 1 s to several minutes, impacting about 6 % of the data. Using a multilayer perceptron with anomaly detection and mitigation strategy improved error detection in testing data, with false positive and false negative rates of 1 % and 2 %, respectively. In the early 2010s, Sayers et al. (2014) utilized multi-objective algorithms to optimize flood risk intervention strategies within an urban drainage system. The original expected annual damage runs took about five and a half hours each. The solutions evaluated using the rainfall set in the third test only took around 10 to 15 min, which was a substantial time saver. In the mid-2010s, Torregrossa et al. (2016) proposed a methodology for conducting daily benchmark analysis on Wastewater Treatment Plants (WWTPs) under database limitations. The authors focused on various methodologies such as fuzzy logic (FL), Artificial Neural Network (ANN), Support Vector Regression (SVR), and Random Forest (RF). The Random Forest algorithm performed the best based on the threshold value of R^2.

2.2. The present state of AI applications in the environmental disciplines

The present state of AI applications in environmental fields is the focus of our exploration in the following paragraphs. Our attention will be directed towards the fields where AI tools are regularly employed, as illustrated in Fig. 1.

2.2.1. Environmental quality monitoring

A concise overview of the various AI applications within the environmental disciplines is provided in Table 2. Tools available for environmental monitoring can analyze air and water quality data and other environmental factors. These tools utilize machine learning algorithms to identify data patterns and predict future trends (Haupt et al., 2009). For example, IoT (Internet of Things) devices like Air Quality Multisensors Systems (AQMS) can provide relatively accurate estimations of air pollutants. These devices deploy pervasive Air Quality Monitoring (AQM) networks that address the issue of geographical sparseness in the current network of AQ Regulatory Monitoring Systems (AQRMS), as noted by De Vito et al. (2020). The ML approach overcomes the limitations of classical pollutant dispersion models with incomplete descriptions of the atmospheric boundary layer hydrodynamics and partial characterizations of the emission inventories. Fabregat et al. (2022) used a model to estimate the individual contribution of different pollutants (CO, NO_x, O_3, $PM_{2.5}$, PM_{10}, and SO_2) from various sources and road traffic restrictions. To train the model, they used air quality measurements collected during the COVID-19 pandemic lockdown when pollutant values were unusually low due to reduced traffic. The model's predictive performance for each pollutant, as measured by R^2, ranged between 0.58 and 0.82. Their research showed that it is possible to accurately measure the individual contributions to increased pollutant concentrations in urban areas without requiring detailed atmospheric hydrodynamics or precise source inventories (Fabregat et al., 2022). With the rapid advancements in AI technologies, it is now possible to monitor air quality indoors using affordable and compact sensors that provide real-time readings of pollutant levels. This is particularly important in indoor spaces where pollutant levels can suddenly spike (Chojer et al., 2022).

A study done by Fijani et al. (2019) developed a hybrid model with an extreme ML framework to monitor water quality parameters like Chlorophyl-a and Dissolved Oxygen in real-time. The framework proved reliable in identifying and analyzing critical water quality parameters. This model is beneficial when handling high-frequency and complex datasets, reducing mean square and mean absolute error by 14.04 % and 7.12 % for Chlorophyl-a estimation and 5.33 % and 4.30 % for Dissolved Oxygen estimation. The study also found that the model could predict the minimum and maximum concentrations of Chlorophyl-a and DO (Fijani et al., 2019). Saboe et al. (2021) conducted a study on microbial potentiometric sensor (MPS) signal patterns that were utilized along with machine learning (ML) tools to predict changes in monitored water quality parameters. The ML tools used were Long short-term memory (LSTM) neural networks for multivariate single-step and multivariate multi-step time series. The difference between measured and ML-predicted water quality parameters in the DO sensor results and MPS-predicted values was less than 10 %. Despite this slight difference, the MPS signals processed by ML tools can be effectively utilized to monitor algae concentrations (Saboe et al., 2021).

To monitor the ocean, AI-powered tools can track water quality, observe ocean temperatures, identify marine life, and detect pollution (Haupt et al., 2009). Automated sensors are connected wirelessly to data stations, visualization tools, and acoustic and physical tagging technologies, which help detect environmental stress and the behavioral changes it causes in fish. The collected data can be used for computer modeling to identify environmental trends, which is vital for predicting some of the effects of climate change. Using AI technologies is essential in gathering and evaluating greater volumes of data required to comply with environmental regulation requirements (Bell et al., 2022). At Ocean Networks Canada (ONC) and other deep-sea cabled observatories, a vast amount of data is collected annually. To effectively manage this data, AI monitoring programs are essential for animal tracking and classification. Bonofiglio et al. (2022) introduced a machine-learning and computer-vision automated pipeline to detect key Pacific Ocean fish species. They employed a YOLO neural network (Deep Learning) to train an automatic detector with a 92 % Average Precision (AP). The detector was then utilized on video materials, and a tracking algorithm was employed to count individual fish. These proxy abundance estimates are among the first to be produced using a large volume of video data from deep-sea settings (Bonofiglio et al., 2022).

2.2.2. Energy analysis

One way to enhance energy efficiency and reduce greenhouse gas emissions is by using AI tools. To achieve more accurate ML models and algorithms, it is crucial to have real-world datasets. In Canada, a pre-cleaned real-house dataset called the Almanac of Minutely Power dataset Version 2 (AMPds2) is accessible to researchers, engineers, scientists, technologists, and utility companies for testing their ML models on energy analysis (Makonin et al., 2016). Ramos et al. (2022) used machine learning (ML) models with real energy data from photovoltaic (PV) systems that can lead to consistent and highly accurate predictions, with a forecasting accuracy of approximately 93–95 % for Generated-Exported prediction (Ramos et al., 2022). In addition, ML methods have proven useful for accurately forecasting electricity generation from renewable energy sources (Krechowicz et al., 2022). For instance, an ML model designed for predicting electricity generation from a PV system achieved an R^2 of 0.94 for short-term forecasting (Pombo et al., 2022). Similarly, a hybrid physical process incorporating artificial neural networks (ANNs) for power prediction in wind turbines achieved an R^2 of 0.91 and surpassed a purely physical model (Zhou et al., 2022).

A recently proposed model for predicting hydropower capacity, which utilized various ML algorithms with different kernels, has proven highly effective. This model outperformed other models and demonstrated a Newman-Pearson correlation coefficient of 0.97 for hydropower plants located in Northern Italy with a total installed capacity of 12.40 TWh (Condemi et al., 2021). In another study, Manfren et al. (2022) tested a data-driven building energy modeling system on a Passive House building. They discovered that modeling data obtained from thermal energy sources displayed high statistical indicators, such as an R^2 of 0.99 for heating and 0.97 for cooling, compared to the dataset from electrical energy sources, which had an R^2 of 0.92 for total. The authors emphasized that ML techniques are powerful and interpretable tools. However, involving human experts in the process and creating feedback loops is essential to continuously improve efficiency measures in Passive House buildings (Manfren et al., 2022). Predicting building energy using ML has certain limitations due to inadequate training data. These limitations can be overcome by identifying and categorizing different situations and incorporating new and innovative sources to improve the performance of the ML model. In the study of Miller et al. (2022), ML models had errors within an acceptable range (below 0.1) on 79.1 % of their test data during testing. Errors of a lower magnitude (between 0.1 and 0.3) occurred in 16.1 % of the test data. Higher magnitude errors (over 0.3) were present in 4.8 % of the test data and are unlikely to be accurately predicted. Including new sources like field data and web-based sources may enhance the effectiveness of ML models in situations where making precise predictions is difficult (Miller et al., 2022).

2.2.3. Environmental modeling and weather pattern predictions

In a recent study by Migallón et al. (2022), environmental modeling was improved using parallel nonlinear models. Real meteorological data from Andalusia, an autonomous community in southern Spain, was used to model the daily maximum air temperature using various methods. The results showed that their nonlinear methodology was effective, with goodness-of-fit measures (R^2 of 0.83 and root mean square error (RMSE) of 3.01) that outperformed other machine learning models like SVM

Table 2
A summary of the discussed studies in Section 2, AI applications in the environmental disciplines. n.a. refers to not applicable or data not found.

Targeted environmental field	AI tools	Performance metrics	Processing time	Best performance	Publication
Air quality monitoring	Shallow Neural Networks (SNN), Extreme Learning Machine (ELM)	Mean absolute error (MAE), mean relative error, normalized MAE, root mean square error (RMSE), nRMSE	4 weeks calibration set, 4 weeks offline training	SNN	De Vito et al., 2020
Air quality prediction	Multilayer Perceptron Regressor (MLPR)	Correlation coefficient (R^2), factor of two of observations (FAC2), geometric mean bias (GeoMean), geometric standard deviation (GeoSTD), RMSE, mean bias	2 h training CPU (Central Processing Unit)	n.a.	Fabregat et al., 2022
Indoor air particulate matter (PM) monitoring	Multiple Linear Regression (MLR), Support Vector Machine (SVM), Gradient Boosting Regression (GBR), Extreme Gradient Boosting (XGB)	R^2, RMSE, mean bias error (MBE)	n.a.	SVM	Chojer et al., 2022
Real-time monitoring of water quality	Least Square SVM, complete ensemble empirical mode decomposition algorithm with adaptive noise CEEMDAN, Variational Mode Decomposition (VMD), ELM, and their combination	R^2, RMSE, MAE, normalized RMSE, normalized MAE, bias error	n.a.	VMD-CEEMDAN-ELM	Fijani et al., 2019
Real-time monitoring and prediction of water quality	Long short-term memory (LSTM) neural network for multivariate single-step and multivariate multi-step time series forecasting	RMSE, normalized RMSE	n.a.	n.a.	Saboe et al., 2021
Ocean monitoring: animal tracking	YOLOv5 (deep learning)	Average precision (AP)	Reduced video processing time from 20 min to 3 min	n.a	Bonofiglio et al., 2022
Energy prediction	Random Forest, SVM, Gradient Tree Boosting (GTB)	MAE, RMSE, mean absolute percentage error (MAPE), prediction accuracy	n.a.	GTB	Ramos et al., 2022
Solar photovoltaic (PV) power forecasting	Random Forest, SVM, Artificial Neural Networks (ANN)	MAE, RMSE, residual sum of squares (RSS), R^2	n.a.	Random Forest	Pombo et al., 2022
Power prediction of wind turbines	Deep Neural Network, Transfer Learning	RMSE, MAE, MAPE, R^2	n.a.	Physical Guided Neural Network (PGNN)	Zhou et al., 2022
Hydro-power production capacity prediction	Multilayer perceptron (MLP), ELM, SVM	RMSE, MAE, correlation	n.a.	SVM	Condemi et al., 2021
Building energy modeling	ML-based regression model	R^2, MAPE, normalized mean bias error (NMBE)	n.a.	n.a.	Manfren et al., 2022
Building energy prediction	Neural Networks, XGBoost	RMSE, MBE	n.a.	n.a.	Miller et al., 2022
Environmental modeling	MLR, K-nearest neighbor (KNN), ANN, SVM	Relative error, R^2, MSE, RMSE, MAE, MAPE	n.a.	SVM	Migallón et al., 2022
Hydrological modeling	LSTM, transfer learning	Nash–Sutcliffe efficiency (NSE)	n.a.	LSTM	Natel de Moura et al., 2020
Solar radiation prediction	Multivariate Adaptive Regression Spline (MARS), SVM, XGBoost, polynomial Ridge Regression (RR)	R^2, RMSE, MAE, MAPE, mean absolute deviation (MAD), uncertainty	n.a.	All of them	Da Costa Alves Basílio et al., 2022
Climate models	Random Forest	R^2, MAE, average test anomaly correlation coefficient (ACC)	n.a.	n.a.	Nichol et al., 2021
Invasive plant species	SVM, Boosted Regression Trees (BRT)	Area under the curve (AUC), RMSE	n.a.	BRT	Sittaro et al., 2023
Wildlife conservation	Computer Vision, Bayesian estimation, Decision Tree, Random Forest, SMV, Deep learning: ANN, CNN, Vision Transformers, LSTM, Gated Recurrent Unit (GRU)	For ML: n.a. For deep learning: accuracy, recall, precision	n.a.	DL algorithms: CNN, Vision Transformers, LSTM, GRU	Tuia et al., 2022
Nature conservation	Decision Tree Classifier (DT), Extra Tree Classifier (ET)	Accuracy, precision, recall, F-score	n.a.	ET	Moran et al., 2017
Energy management on domestic hot water	ANN	R^2	n.a.	n.a.	Maltais and Gosselin, 2022
Water management, groundwater modeling	Random Forest, SVM, and deep learning: ANN	n.a.	n.a.	Random Forest	Miro et al., 2021
Hydrological drought forecasting	Deep learning: ANN and ML: SVM	Confusion matrix	n.a.	All of them	Almikaeel et al., 2022
Waste management	Conditional Random Forest, Univariate Non-linear Regression	RMSE, symmetric mean absolute percentage error (SMAPE), Akaike information criteria corrected (AICc)	n.a.	All of them provided different insights	Velis et al., 2023
Waste management	Rough Set-based Machine Learning (RSML)	Accuracy	n.a.	n.a.	Gue et al., 2022

(*continued on next page*)

Table 2 (*continued*)

Targeted environmental field	AI tools	Performance metrics	Processing time	Best performance	Publication
Waste management	Computer Vision, Deep learning: EfficientDet, YOLOv5	Intersection over union (IoU), precision, recall, average precision (AP)	n.a.	YOLOv5	Moral et al., 2022
Waste management	Computer Vision, Deep learning: Faster-RCNN, RetinaNet, Grid-RCNN, YOLOF, Mask-RCNN, Cascade Mask-RCNN, YOLACT, SOLOv2	Mean average precision (mAP), AP, normalized confusion matrix	n.a.	YOLOF, SOLOv	Agnew et al., 2023

(Support Vector Machine) and ANNs (Artificial neural networks). Furthermore, the training process was more straightforward, as demonstrated in the study (Migallón et al., 2022). Studies have indicated that the traditional approach to forecasting maximum streamflow values is prone to inaccuracies. As an alternative, ML has emerged as a reliable method for predicting daily streamflow. The application of Forward Feature Selection (FFS) to select covariables has resulted in further improvements to the efficacy of ML-based hydrological models, as demonstrated in a study conducted by Reis et al. (2021). Natel de Moura et al. (2020) conducted a study that tested the effectiveness of machine learning (ML) approaches in predicting daily discharge using daily precipitation and temperature data. The researchers also calibrated the approaches in a wet period and validated them in a dry one, and vice versa, to assess climate transposability. They found that using a more complex approach, such as deep learning, resulted in better performance than more straightforward methods. They explained that this is likely due to the ability of the memory cell to simulate storage effects (Natel de Moura et al., 2020).

Advanced software programs called climate modeling tools use complex algorithms to simulate weather patterns and predict potential changes. With the help of AI, these tools can evaluate climate adaptations and provide future weather projections and simulations. By utilizing corrective machine learning (ML) methods, weather forecasts have improved and demonstrated accurate predictions up to 10 days in advance. Utilizing the corrective ML has resulted in a 30 % reduction in time-mean precipitation biases and global time-mean errors compared to traditional climate models (Bretherton et al., 2022). It has been found that machine learning (ML) models can accurately predict solar radiation levels at various sites around the globe. The modeling process can become less complex while improving prediction accuracy by selecting the appropriate environmental variables for input. Furthermore, optimized ML models have been shown to predict global horizontal irradiation with small error rates, ranging between 0.12 and 0.14 Mean Absolute Percentage Error (MAPE), as Da Costa Alves Basílio et al. (2022) reported. Nichol et al. (2021) studied Arctic sea ice extent in September 2020 using machine learning-based Earth system models (ESMs) for climate prediction. The accuracy of these models is crucial for economic, ecological, and geopolitical planning. Previous models have often underestimated Arctic sea ice loss, but their study aimed to improve prediction accuracy. The study showed that the Energy Exascale Earth System Model (E3SM) relies too heavily on just one of the ten selected climatological quantities to predict September sea ice averages. Additionally, E3SM gives too much importance to six of those quantities when compared to observed data. The authors concluded that identifying the correct features can help climatologists improve the prediction accuracy of already existing climate models (Nichol et al., 2021).

2.2.4. Environmental research

Researchers in the environmental disciplines can benefit from using machine learning. AI-powered tools are becoming increasingly important in monitoring and protecting biodiversity. These tools can identify endangered species, track their movements, and detect habitat loss (Corbane et al., 2015). Using ML algorithms, various habitats can be analyzed where invasive plant species have spread and disrupted ecosystem functions. These algorithms can create species distribution models and monitor the environment to identify the factors responsible for plant species invasion under different climate conditions. Additionally, ML algorithms can also assess how invasive species impact habitat suitability. Sittaro et al. (2023) conducted a study that revealed the good predictive ability of ML algorithms, which helped to identify the factors that determine the invasion of plant species. They found that the algorithms had a RMSE measure of 0.23 and 0.29. Regarding wildlife conservation practices, animal ecology can benefit from the vast data generated by modern sensors. By combining ML techniques with their domain knowledge, ML can enhance the accuracy of ecological models and develop integrated hybrid modeling tools. However, this requires interdisciplinary collaboration to ensure the quality of new approaches and to train a new generation of data scientists in ecology and conservation (Tuia et al., 2022). Moran et al. (2017) utilized a classification approach based on machine learning. The method produced classification results that were traceable, interoperable, and observation-based. They tested grassland data obtained in Germany and derived from the European Union Nature Information System (EUNIS). The developed methodology was an initial step towards building ontologies grounded in observation for nature conservation. The assessments indicated encouraging outcomes for identifying grassland indicators such as wetness and alkalinity. The accuracy rate for alkalinity was 85 %, and for wetness, it was 76 %, demonstrating interoperability (Moran et al., 2017).

Optimizing water use and modeling hydrological systems can also be achieved using AI tools. Maltais and Gosselin (2022) assessed Model Predictive Controls (MPCs) that relied on machine learning forecasting models. They simulated 40 different domestic hot water profiles to determine how the MPCs could reduce the energy consumption of Canada's residential sector, which is currently approximately 17 % due to domestic hot water production. The study found that both perfect and imperfect forecasts could lead to a similar performance in most cases, demonstrating the applicability of the ML models. Additionally, the MPC utilizing an ML forecasting model resulted in energy savings ranging from 4 % to 8 % compared to traditional controls (Maltais and Gosselin, 2022). Miro et al. (2021) evaluated the performance and ability of three ML algorithms to simulate the output of a high-resolution MODFLOW model. MODFLOW is a widely recognized hydrologic model developed by the U.S. Geological Survey and used for predicting groundwater conditions and interactions between groundwater and surface water. The authors discovered that using machine learning techniques significantly reduced the time required for modeling and enabled greater exploration of the uncertainty space. This included a wider range of potential future climate changes and drought conditions (Miro et al., 2021). Almikaeel et al. (2022) conducted a study on hydrological drought in the Gidra River in Slovakia. They used two ML algorithms to identify dry, normal, and wet hydrological situations, achieving high accuracy rates in predicting hydrological drought. The study highlights the potential of integrating water management with ML models for irrigation planning and mitigating the effects of drought (Almikaeel et al., 2022).

AI tools help manage waste by optimizing collection routes, identifying and sorting various types of waste, and predicting waste generation patterns. Velis et al. (2023) thoroughly analyzed cities worldwide, focusing on solid waste management practices. They used machine learning and regression analysis to find the best models for explaining the impact of socioeconomic factors on mismanaged municipal solid

waste. Their work helps cities work towards sustainable development goals by establishing a baseline for measuring progress. They found that low-income countries had a median of 45 % controlled recovery and disposal rates, lower-middle-income countries had 75 %, and upper-middle and high-income countries had 100 % (Velis et al., 2023). In the study of Gue et al. (2022), an ML model based on rules was developed to analyze how city and country attributes affect waste management. The model utilized rough set-based ML and was developed using data from 100 cities across 41 countries. The results demonstrated that the rough set-based ML models had binary classification accuracies ranging from 89 % to 91 % (Gue et al., 2022). In a study by Moral et al. (2022) in 11 Spanish cities, Computer Vision algorithms were implemented to detect waste containers and determine their location and size. The authors selected two object detectors based on deep learning techniques that were proven effective and adaptable for real-time usage. The system achieved an average accuracy rate of 89 % (Moral et al., 2022). Agnew et al. (2023) conducted a study where they utilized Computer Vision techniques to automate and improve the process of detecting the status of bins. They successfully achieved a mean average precision (mAP) of 80 % or higher with each model, demonstrating the efficacy of using computer vision to automate the recording of overfilled bins (Agnew et al., 2023).

2.3. The environmental impacts of current AI technology

Although the rapid advancement of Artificial Intelligence (AI) tools made them attractive to users of several research fields, where massive amounts of data must be processed, modern computing systems that run complex AI applications still have a few drawbacks. High-precision computing requires extensive energy consumption (Zador et al., 2023), expenses, and time, making AI an unsustainable and costly option for both the environment and the economy (Boahen, 2022; Mehonic and Kenyon, 2022). Environmental researchers and professionals might raise concerns about whether the benefits of using AI tools outweigh the costs. Meulemeester and Martens (2023) compared common data science to advanced data science and found an exponential power consumption associated with the complexity of data science. Using AI applications on a regular computer consumes 2.57 times more power than other applications. Researchers have spent millions of dollars to develop, improve, and operate AI models within a few weeks. Moreover, the energy-intensive workload required to train AI models can contribute significantly to carbon emissions (Brown et al., 2020), equivalent to the carbon release of an estimated 1300 cars (Boahen, 2022). Training a neural network is energy-intensive, requiring about 1000 MWh to train a large language model. This energy is equivalent to the power needed to run a small town for a day (Patterson et al., 2021). In comparison, biological systems, like the average human brain, use far less energy, consuming only about 20 W (Sokoloff, 1960).

Anthony et al. (2020) estimated 37.445 kWh of electricity to train two deep-learning models on medical images using the open-source tool Carbontracker on Python, contributing to 3.166 kg of carbon emission (CO_2eq). A study by Desislavov et al. (2023) analyzed various elements of AI to understand its impact on energy consumption better. The researchers concluded that AI models can still maintain improved performance through algorithmic enhancements and increased parameters, even if their energy usage remains constant. However, their conclusion has a drawback as it assumes a consistent multiplier. With the increasing integration of AI technology, energy consumption is bound to rise due to higher usage. This trend might look like the rise in the number of cars, despite their improved efficiency over the last two decades. According to Trihinas et al. (2022), the environmental impacts of training AI models can vary based on their location. The device and resource configurations used can affect the performance and power needed for a model's inference. It is crucial to test different configurations to determine which can provide the necessary reliability and performance while minimizing the environmental impacts (Trihinas et al., 2022).

The time at which an AI model is trained can significantly impact the amount of carbon emissions it generates. Trihinas et al. (2022) found that training an AI model during midday results in a carbon footprint of 1.64 kg less than if the training is initiated at 6 pm and 2.3 kg less than if training is initiated at 9 pm in Cyprus. The difference in carbon footprint can be explained due to the energy sources used by the power grid. Trihinas et al. (2022) suggest that organizations that aim to reduce their carbon footprint should train their AI models when low-carbon energy sources are utilized to power the computing system. A study by Dube et al. (2019) found that the runtime for training AI models in the cloud, both online and offline, can take anywhere from 2 to 4 days to complete and is affected by various factors such as the GPU (Graphics Processing Unit), deep learning frameworks, and algorithms used to optimization. Choosing the right AI model architecture is also important since there can be trade-offs to consider. For instance, SqueezeNet has a faster inference time than ResNet50, but its accuracy is more than 20 % lower. On the other hand, MobileNetV2 has a smaller runtime footprint and only a 3 % reduction in accuracy compared to ResNet50. MobileNetV2 also can reduce energy consumption by almost 90 % compared to ResNet50. It is important to consider factors beyond accuracy metrics, such as resource overhead, energy footprint, and environmental impact, when selecting an AI model. Different model architectures can impact system utilization and battery autonomy (Trihinas et al., 2022).

Creating new computational models requires significant power and resources, leading to increased costs. Some models cost up to an estimated $700,000 daily to operate and answer common questions (Jordan, 2020). In a study conducted by Strubell, Ganesh, and McCallum in 2019, various Natural Language Processing (NLP) models were analyzed to determine the environmental and computational costs of training deep learning neural network models for NLP. The study found that using Tensor Processing Units (TPUs) was more cost-effective for power consumption during training than Graphics Processing Units (GPUs). However, even the most sustainable language model still costs at least $150,000 for cloud computing and has a considerable carbon footprint. The most expensive language model had a cloud computing cost ranging between $942,973 and $3,201,722 and a training time of 274,120 h. Strubell et al. (2019) also discovered that while training a single model is inexpensive, adjusting that model and carrying out comprehensive R&D (research and development) to improve the model quickly escalates in cost in terms of time, processing, and power usage (Strubell et al., 2019). Researchers invested $4.6 million and ran 9200 GPUs for two weeks to train a well-known language model (Boahen, 2022). Governments are investing in quantum technologies by collaborating with industry and academia to enhance the development of sensors and metrology, imaging, communications, and computing. For instance, the UK has allocated roughly £1 billion (approximately $1.24 billion) towards quantum technologies through its National Quantum Technologies Programme. China has established a multi-billion-dollar National Laboratory for Quantum Information Sciences, and the USA has commissioned a National Strategic Overview for Quantum Information Science, resulting in a $1.2 billion investment over five years (Mehonic and Kenyon, 2022).

2.4. AI and environmental ethics

The study of ethical relationships between humans and the natural environment is known as environmental ethics. It encompasses societal attitudes, actions, and policies that aim to protect and sustain biodiversity and ecological systems (Attfield, 2014). Artificial intelligence (AI) can be useful in addressing ethical concerns in this field. By analyzing new situations and providing guidance to environmental professionals, AI can help ensure that actions taken are appropriate and ethical (Thomson, 1997). However, some environmental professionals argue that AI ethics have limited and inconsistent attention to “non-human” biological species. It is crucial to consider the moral treatment of every species in AI ethics research and practice, as it affects the

content of AI ethics principles and the choices AI makes. It is also important to monitor the unintended consequences of AI systems, such as their use of resources and energy. Incorporating the interests and values of every species into AI system design is essential (Owe and Baum, 2021).

Speciesism refers to the belief in human superiority, which often results in the mistreatment and exploitation of animals (Ryder, 2011). Hagendorff et al. (2022) found that mainstream AI applications, particularly in computer vision and natural language processing, reinforce biases related to speciesism. This is because these models are trained on datasets that already have existing speciesism patterns. Consequently, when machine learning algorithms are trained based on past behavior, biases such as historical, position, exposure, and popularity biases become more likely to occur in recommender systems (Hagendorff et al., 2022). There are ongoing efforts to reduce biases in data and algorithms to ensure AI applications are fair in the environmental fields. Mäntymäki et al. (2022) presented a comprehensive AI governance modeling framework, known as the hourglass model of organizational AI governance, which can assist organizations already using AI systems to implement ethical AI principles in practice. Both environmental and internal organizational layers are integrated into the AI systems. This model serves as a starting point for decision-makers to consider the governance components necessary for ensuring social acceptability, mitigating risks, and maximizing the potential of AI.

2.5. Environmental implications of the social effects of AI

Decision tools for land, water, and environmental management should be analytical and consider physical, biological, socioeconomic, and policy factors. They should also be user-friendly and applicable to the needs of the clients (Cortés et al., 2000). End-users have concerns about the reliability of AI methods, which makes it crucial for creators to prioritize building trust with them. It is essential to examine and determine ways to enhance trust in AI tools from a social perspective (Haupt et al., 2022). The quality of information available plays a significant role in protecting the environment effectively. Problems arise when there is an overwhelming amount of non-uniform information, and it is difficult to determine its quality beforehand (Cortés et al., 2000). For instance, there is a lot of information available in every field, but sometimes, it can be flawed, leading to misinformation due to accidental or intentional biases. For complex environmental issues such as environmental racism, climate change, global environmental change, biodiversity loss, or pollution, these biases can be perpetuated and amplified by the training data used in large language models. Conversely, people who already have easy access to environmental information could benefit even more from these tools. The availability of large language models in multiple languages could also broaden the participation of people in environmental discussions, as suggested by Rillig et al. (2023). Cooperation between environmental and AI professionals is necessary to eliminate biases and manage environmental solutions in a decentralized manner on local and global levels (Cortés et al., 2000).

3. Advantages of using AI tools

The introduction of AI tools into the workplace has several advantages, including the creation of new industrial processes, increased productivity, and a redefinition of the position of human workers who will now supervise the task of AI tools. Using AI technologies and procedures promises to lessen human workers' danger, monotony, and physical exhaustion (CORDIS EU, 2020). Digital tools are not replacements for human experts but help to enhance information gathering for human experts and promote interdisciplinary perspectives (Grubert and Siders, 2016). AI tools can handle large amounts of environmental data and identify patterns that may be difficult for humans to detect manually. It can lead to more accurate predictions and recommendations, as well as faster and more effective data processing and analysis (Haupt et al., 2022). AI algorithms can monitor and identify environmental issues like air pollution, water quality problems, and natural disasters in real time. For instance, neural network models can map air pollutants and air quality health indices in real-time with mobile monitoring units, providing more coverage than classic stationary monitoring system units (Adams and Kanaroglou, 2015). Adams and Kanaroglou (2015) found that classic stationary monitoring system units could only map 1.45 % of the total region that mobile monitoring systems could do with neural network models involved (Adams and Kanaroglou, 2015).

Using the advantages of AI systems can also provide early detection of environmental issues, but also can help reduce damage and minimize the risk of harm to human health. AI systems can improve performance, computational efficiency, and modeling flexibility by using machine learning to model complex and non-linear systems like the environment. This approach has also allowed for discovering of new information about non-linear systems, including the factors contributing to severe events (Haupt et al., 2022). Vitanza et al. (2022) used a multi-model machine-learning method to detect extreme rainfall events in Sicily and identify weather anomalies. These events can occur suddenly and cause significant social, environmental, and economic consequences, as seen in the 2021 rainfall that led to 300 mm of rain near Catania, Sicily. Their approach helped to identify areas in Sicily that could experience future extreme events similar to the one in 2021 (Vitanza et al., 2022).

Using AI tools can help to identify innovative solutions for environmental challenges by analyzing data from various sources and detecting patterns that may not be visible to humans. Using AI tools can lead to more effective decision-making in environmental issues and advancements in the field (Hino et al., 2018). Recently, the European Commission established the “European High-Performance Computing Joint Undertaking” (EuroHPC JU) project to develop a world-class European supercomputing ecosystem (Table 3). The EuroHPC JU project is focused on creating a state-of-the-art supercomputing ecosystem in Europe to improve competitiveness, promote innovation, advance science, and enhance the quality of life for European citizens. A key objective of the project is to support carbon neutrality goals outlined in the European Green Deal by promoting green and sustainable technologies. To achieve this goal, the EuroHPC JU is building supercomputers by utilizing water cooling, waste heat recycling, and energy-efficient microprocessors (European Commission, 2022). Supercomputers are incredibly powerful machines that can solve complex problems and perform calculations beyond the capabilities of traditional computers (Krause and Thörnig, 2018). Supercomputers are crucial in areas like climate modeling and energy-saving strategies, making them essential for anticipating severe weather conditions, developing sustainable products, and much more (European Commission, 2022).

The use of AI and machine and deep learning is crucial in designing, modeling, and automating repetitive tasks. With the help of machine learning and deep learning, AI applications can quickly learn from data and results, analyze new information from multiple sources, and adapt with great accuracy, providing significant value to scientists, engineers, and businesses (Haupt et al., 2009). In a study conducted by Kadow et al. (2020), they utilized image inpainting techniques to fill in missing data points in gridded spatial climate datasets. Their approach involved using stacked partial convolutions and an automatic mask updating mechanism through transfer learning to reconstruct the missing values (Kadow et al., 2020). Deep learning algorithms are ideal for predicting spatial stochastic coefficients because they can automatically learn feature representations. Amato et al. (2020) used a non-linear model that combined temporal bases and spatial coefficient maps related to the environment and climate, using a single deep learning structure with multiple outputs to model all spatial coefficient maps coherently (Amato et al., 2020).

Efficient resource management is essential for a sustainable future, and artificial intelligence (AI) systems are playing a crucial role in

Table 3
Example projects involved in the European supercomputing ecosystem (EuroHPCU JU) program that prioritize environmental and sustainable goals.

Project name	Main goal	Contribution to the scientific fields	Country	Publications
JURECA	Quantum computing	Climate, earth, environment, and more	Germany	Krause and Thörnig, 2018
MaX	Quantum simulations of materials	Information technology, green energy, health	Italy	Zaccaria et al., 2020
MultiXscale	Software development	Civil transport, battery applications for sustainable energy transition, and more	Slovenia	No publications available yet
HiDALGO2	Applications development for simulations	Urban air quality, energy efficiency of buildings, renewables, wildfires, and more	Poland	No publication available yet
ChEESE-2P	Modeling and simulations	Earth sciences	Spain	Titos et al., 2022
ESIWACE3	Scalable simulations	Earth systems modeling, weather, and climate predictions	Spain	Stephan et al., 2022
eProcessor	Open-source European full stack ecosystem	European ecosystem	Spain	Bacardit et al., 2021
MAELSTROM	Weather and climate modeling	Future effects of climate change	United Kingdom	Meyer et al., 2022
ACROSS	Boosting simulations' speed with machine and deep learning	Global numerical weather predictions, climate simulations	Italy	Willner et al., 2022

achieving this goal. By accurately identifying areas where water, energy, and land are underutilized, AI technology helps optimize resource management and minimizes negative environmental impacts. Strubelt et al. (2018) used a simulation model to improve logistics processes at an underground waste storage site in Germany, reducing setup times and scheduling issues. Moreover, AI systems automate data processing and analysis tasks, making environmental monitoring and analysis more accessible and affordable for smaller organizations (Haupt et al., 2009). Already, AI is helping with pollution control, environmental management, and industrial waste management worldwide, and it is expected to play an essential role in achieving sustainable development goals (Ong et al., 2022).

4. Current challenges and limitations

While AI tools and ML models have proven to be advanced technologies in various scientific areas, they have faced difficulties and boundaries in the environmental sector. These challenges have resulted in unsatisfactory outcomes and the inability to gain approval from environmental experts. A major obstacle to achieving satisfactory results in real-world scenarios is the need for a large amount of labeled data during training. Machine learning models are capable of capturing complex spatiotemporal relationships, the requirement for a significant amount of labeled data remains a significant challenge (Willard et al., 2022). It is common for environmental data to have missing information due to technical errors during data collection. Examples of such errors include a sensor or weather station running out of batteries or a power outage occurring during environmental data collection. A study by Amato et al. (2020) explored the potential of ML techniques to address the issue of missing data in environmental time series. In forecasting scenarios, it is typical to view the various measurement locations as nodes in a graph and utilize specialized DL architectures to model the correlations among them. However, this methodology has a significant drawback: predictions can only be made at the spatial locations of the measurement stations and not at other locations that may be of interest (Amato et al., 2020). Amato et al. (2020) proposed that additional research should be conducted to create methods for accurately measuring the spread of uncertainty throughout all aspects of their proposed DL framework for spatiotemporal climate and environmental data prediction.

Processing environmental data can also be challenging, especially when analyzing time-series patterns to predict future trends. Researchers have been improving time series modeling and forecasting accuracy and efficiency. ML and DL algorithms can be difficult to use if the data is noisy or error-prone. This can lead to models struggling to capture essential features during exploration. It is crucial to filter and denoise the data to ensure a good model fit, making it easier to explore, understand, and extract features (Haq et al., 2022). Although current denoising algorithms have limitations, a new approach involves using an increasingly accurate understanding of physics to remove noise from signals. Traditional algorithms may struggle to program complex real-world scenarios accurately, but deep learning techniques can improve signal-to-noise for various applications (Ede, 2020). Another issue in the environmental discipline is ensuring that water resources are of good quality, which can be difficult to maintain due to the complex mixture of chemical contamination from various sources and the harmful biological effects of these contaminants. The challenge is compounded when regulatory efforts focus on monitoring only a few chemicals, creating tensions between experimentalists, water quality modelers, and stakeholders. As a result, this can lead to problems with data collection and sharing, difficulties in publishing reproducible model workflows, and a lack of expertise in using high-performance computing systems for reliable experimental simulations. Additionally, adapting loosely coupled or component-based models is necessary for developing reliable software implementations of integrated modeling frameworks (Fu et al., 2020).

To effectively utilize AI to understand a system better, scientists must possess a solid understanding of its physics. Key factors in configuring AI methods like machine learning include selecting predictive features, informing model architecture, grouping physically correlated features, and enforcing constraints to ensure model calibration produces physically consistent solutions (Haupt et al., 2022). When employing deep learning, it is important to acknowledge its limitations. One such limitation is that DNNs (Deep Neural Networks) are vulnerable to adversarial attacks, which can result in drastic errors even with minor input changes. There are ways to improve the training process to bolster resistance to these attacks. Additionally, working with ANNs (Artificial Neural Networks) can be difficult because of unclear code dependencies, pre-trained models, and hardware accelerators. To circumvent these hurdles, it is advisable to use cloud containers to facilitate ANNs (Ede, 2020). A supercomputer operated in Germany was designed to serve as a versatile scientific instrument for compute- and data-intense science, such as simulation. Environmental scientists and engineers interested in using the supercomputer's capacities and capabilities must demonstrate that they have an appropriate knowledge of high-performance computing (Krause and Thörnig, 2018).

Machine learning has been criticized by many forecasters and end users due to its alleged “black box” nature, making it difficult for users to comprehend the rationale behind its predictions. ML practitioners and users are actively striving to address this issue and enhance the

interpretability of machine learning (McGovern et al., 2019). McGovern et al. (2019) emphasized that understanding the balance between predictability and interpretability is essential. ML models do not necessarily model a physical problem but may discover more optimal solutions. To completely replace physically based numerical weather and climate models with AI/machine learning models has limited acceptance among researchers. This is because of their inability to provide a meaningful physical interpretation of underlying processes, their large data requirements, and their limited generalizability to out-of-sample scenarios. Given that neither an ML-only nor a physically based–only approach can be considered sufficient for complex scientific and engineering applications, the research community has been exploring the continuum of hybrids of physically based and ML-based models, where both scientific knowledge and data are integrated synergistically (Haupt et al., 2022). It is crucial to have a strong foundation in proper statistical analysis methods to utilize AI for environmental fields effectively. This includes appropriate training and testing methods and comparing accuracy to other techniques. Collaboration and advancements in training and supporting new AI practitioners are also important. AI specialists should develop guidelines for model development and usage and prioritize the understandability, explainability, and trustworthiness of AI for stakeholders. Additionally, incorporating physics-based constraints can improve the interpretability of results and enhance model performance, as shown by McGovern et al. (2019).

AI methods still heavily rely on human expertise (Ede, 2020). For instance, manually labeling data can be costly, time-consuming, or unsuitable for sensitive data. To overcome these challenges, unsupervised semantic segmentation can be utilized, which involves learning to segment image data from an additional dataset of segmented images or image-level labels. Unsupervised semantic segmentation networks tend to be less precise than supervised networks. AutoML is gaining popularity because it can outperform human developers and potentially save time and cost. Its current limitations are that it can only work with established Artificial Neural Network architectures and learning policies (Ede, 2020). However, sometimes human biases can introduce errors when AI systems are designed for one context, such as ecological, climate, or social-ecological, and then are transferred to another. Although the training data and model may have been suitable for the initial social-ecological scenario, applying it in a different setting could lead to inaccurate and harmful outcomes. Another issue is when AI systems based on past ecological conditions are prone to fail when ecosystems change unexpectedly and irreversibly. Furthermore, an AI system can function as intended by its creator, but the user may not comprehend its usefulness or try to extract meanings beyond its capacity (Galaz et al., 2021). The obstacles and restrictions demonstrate the importance of thoughtful analysis and teamwork between professionals in AI and environmental fields to tackle the urgent environmental issues we face today. Developing AI tools that can monitor near real-time the environmental status, model real-world scenarios, and train environmental professionals remains a future task.

5. Policy and practical implications of the present review

Efforts are currently underway to regulate AI and establish the first comprehensive AI law. In 2023, both the European Union (EU; European Parliament, 2023) and the United States of America (National Conference of State Legislatures, 2023) are working to legislate AI systems that are safe, transparent, traceable, non-discriminatory, and environmentally friendly to prevent harmful outcomes. The Members of the European Parliament (MEPs) adopted the "Parliaments negotiating position on the AI Act" in June 2023, with the goal of reaching an agreement by the end of the year. They also published a proposal for EU legislation known as the "Artificial Intelligence Act"[1] (European Parliament, 2023). Similarly, the United States held a legislative session in 2023 through the National Conference of State Legislatures to introduce bills that examine and develop federal standards to prevent the potential misuse and unintended consequences of AI. The National Institute of Standards and Technology held workshops and discussions with the public and private sectors to develop federal standards for creating reliable, robust, and trustworthy AI systems (National Conference of State Legislatures, 2023).

The topic of Artificial Intelligence (AI) is currently receiving a lot of attention, especially in 2023. It is present in both our personal and professional lives, and because it is developing so rapidly, it is difficult to cover all aspects of it in a single review. Nevertheless, we endeavored to discuss as much as possible in our review, focusing on recent applications of AI in environmental disciplines, ensuring its originality and uniqueness. We hope our aim was fulfilled to inspire readers, environmental professionals, and AI experts to discuss and collaboratively solve environmental issues for mutual benefit.

6. Conclusion

We reviewed the latest applications of Artificial Intelligence (AI) tools in the environmental discipline. We explored the opportunities that AI offers and discussed the advantages and disadvantages of implementing these tools in the environmental fields. AI technologies are helping to manage and assess large amounts of environmental data from sources such as demographics, traffic trends, and energy usage. Integrating big data and AI offers new possibilities for environmental tasks such as modeling, monitoring, and research. There are ongoing efforts to develop AI tools to address various environmental challenges, with the potential to improve environmental management and sustainability. Environmental professionals began incorporating AI tools into their work in the early 2000s, but usage increased exponentially after 2010. There has been an increase in the use of AI applications in environmental science and engineering, particularly for modeling, monitoring, energy analysis, research, and predicting climate change.

AI-powered tools allow for real-time monitoring of air and water quality. Machine learning (ML) algorithms can predict future trends and critical parameters and accurately detect key fish species. IoT (Internet of Things) tools provide accurate estimations and overcome geographical limitations. Deep-sea cabled observatories gather vast amounts of data managed using AI monitoring programs. The use of AI tools has the potential to enhance energy efficiency and decrease emissions. However, to develop accurate ML models, it is necessary to have access to real-world datasets. When working with genuine energy data, ML models can accurately forecast electricity generation from renewable sources. For example, for PV (photovoltaic) systems, the R^2 is 0.94, while for wind turbines, it is 0.91. Recent studies demonstrate that ML algorithms can also make reasonably accurate predictions about hydropower capacity and building energy modeling. Incorporating additional sources of data can further enhance the effectiveness of ML models. Forecasting maximum streamflow values and predicting solar radiation levels can be done reliably using ML. Deep learning (DL) has proven to be more effective for daily discharge prediction than simpler methods. Climate modeling tools powered by AI can analyze climate change indicators and predict potential changes, but accuracy can be improved by identifying the correct features for analysis. ML can help protect biodiversity by identifying endangered species, tracking their movements, and detecting habitat loss. It can also optimize water usage, reduce energy consumption, and improve waste management models. Collaboration between ecology and data science is crucial for effective conservation efforts.

[1] https://www.europarl.europa.eu/RegData/etudes/BRIE/2021/698792/EPRS_BRI(2021)698792_EN.pdf.

Advanced technology, such as supercomputers and AI tools, plays a vital role in predicting weather, creating environmentally sustainable products, and automating tasks. These tools enable engineers, scientists, and businesses to analyze data and improve resource management, reducing waste and lessening environmental impact. Artificial Intelligence also assists in pollution control and helps achieve environmental sustainability goals. However, developing and utilizing AI tools can cost tens of millions of dollars and require extensive time and energy-intensive workloads. This high cost can limit access to these tools for many organizations and individuals who could benefit from them, resulting in disparities in environmental data and insights among different stakeholders and regions. Additionally, the cost of Research and Development for AI is influenced by the complexity of the technology and the level of innovation. Furthermore, obtaining and preparing high-quality datasets for specialized domains can be costly.

AI applications face challenges in the environmental sector due to the need for a large amount of quality data, the fear of data sharing, missing information, and noisy data. ML techniques can improve time series modeling and forecasting, but data must be filtered and denoised. In order to use AI to gain a deeper understanding of a system, scientists need to have a firm grasp of its underlying physics. Many modelers and end users have criticized machine learning for its "black box" nature, which makes it challenging for users to understand the reasoning behind its predictions. AI techniques still heavily depend on human expertise, but unsupervised machine learning or transfer learning can be used that learns from a dataset. However, unsupervised networks are typically less accurate than supervised ones. AutoML is becoming more popular because it can surpass human developers and has the potential to be more effective. Collaboration and advancements in training new AI practitioners in the environmental disciplines are important.

AI technologies can improve environmental data processing, assessment, and decision-making by enhancing precision, efficacy, and efficiency. Environmental data and resources can be better utilized by leveraging the power of AI to preserve the well-being of our planet and its inhabitants. However, it is important to exercise caution when applying AI tools in the environmental field, as there may be limitations and unintended consequences. Environmental professionals should evaluate the advantages and disadvantages of utilizing AI tools to ensure that they make accurate, environmentally sustainable, and effective decisions and data assessments. It is not right to judge AI and its tools based solely on their present limitations since they are still in the early stages of development and are continuously improving. The full potential and advantages of AI are yet to be realized, and the creation of low energy-consuming systems that can process and store data more efficiently is on the way (Strubell et al., 2019; Trihinas et al., 2022; European Commission, 2022; Desislavov et al., 2023). Furthermore, ongoing research is underway to evaluate the environmental impact of AI development and its optimal usage in the environmental field.

CRediT authorship contribution statement

Aniko Konya: Conceptualization, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **Peyman Nematzadeh:** Conceptualization, Investigation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors have no financial or non-financial competing interests related to this work.

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